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46310 Wind Turbine Aeroelasticity

Derivation of the equations of motion for a wind turbine for course 46310 at DTU

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1 Overview

In this document, a step-by-step derivation of the equations of motion for the simplified wind turbine model of course 46310 at DTU is presented. Section 2 introduces generalised coordinates and forces – two backbones to the subsequent derivations. Section 3 derives the equations of motion for a single wind turbine blade (a beam) that is then extended to the full wind turbine in Section 4.

2 Generalised Coordinates and Forces

This section first introduces the generalised coordinates to derive the generalised forces. Small examples and exercises are given throughout and at the end.

Using Lagrange's equations, we are not restricted to our *common* Cartesian coordinate system. Instead, we can choose any suitable coordinate system and call the coordinates *generalised* coordinates. The relationships between our *common* coordinates x_{ij} and the generalised coordinates q_i are given as (Goldstein et al., 2002)

$$x_{ij} = f_{ij}(q_1(t), q_2(t), \dots, q_n(t), t) \quad (1)$$

where n is the number of generalised coordinates, i is the particle index, and j is the axis index.

Example: Polar Coordinates Take as an example the description of a single particle ($i = \{1\}$) in 2D space ($j = \{1, 2\}$) using polar coordinates r, φ . If we now say that axis $j = 1$ is x and $j = 2$ is y , we get

$$\begin{aligned} x_{11} &= x_1 = r \cos(\varphi) \\ x_{12} &= y_1 = r \sin(\varphi) \end{aligned} \quad (2)$$

With that we have

$$\begin{aligned} f_{11}(r, \varphi) &= f_{1x} = r \cos(\varphi) \\ f_{12}(r, \varphi) &= f_{1y} = r \sin(\varphi) \end{aligned} \quad (3)$$

So, our generalised coordinates in this example are simply the polar coordinates r and φ .

In general, we want to come up with as few generalised coordinates as are necessary to describe the displacement of our system. The transformation f_{ij} may or may not depend *explicitly* on time t . When f_{ij} does not depend on t explicitly, like in the example above, we simply write $f_{ij}(q_1, \dots, q_n)$. $q_i = q_i(t)$ generally still applies though!

Now we introduce virtual displacements as infinitesimal, instantaneous, hypothetical displacements consistent with the constraints (e.g. the clamp constraint of a clamped beam). The main difference between a virtual displacement δx and a differential displacement ∂x is that the latter happens over a non-zero time dt . Notation-wise, we use δ only for virtual quantities and ∂ for partial derivatives.

Using basic differentiation rules, it follows that the virtual displacement of x_{ij} is

$$\delta x_{ij} = \frac{\partial f_{ij}}{\partial q_1} \delta q_1 + \frac{\partial f_{ij}}{\partial q_2} \delta q_2 + \cdots + \frac{\partial f_{ij}}{\partial q_n} \delta q_n = \sum_k \frac{\partial f_{ij}}{\partial q_k} \delta q_k \quad (4)$$

where k iterates over all generalised coordinates. Since the virtual displacement is instant ($\delta t = 0$), the time derivative does not appear in δx_{ij} . In words, the summands of Eq. (4) specify how much degree of freedom ij virtually moves when degree of freedom q_k is virtually displaced.

If there are forces present in the system, we may look at their virtual work done for such a virtual displacement. Why would we be interested in that? Because Lagrange's equations are built on the principle of virtual work, which the reader may read up on themselves. Introducing a force acting on particle i in direction j as X_{ij} , we calculate the virtual work done by all forces when all particles are virtually displaced in all directions as

$$\delta W = \sum_i \sum_j \delta x_{ij} X_{ij} \quad (5)$$

Inserting the expression for δx_{ij} from above yields

$$\delta W = \sum_i \sum_j \sum_k \frac{\partial f_{ij}}{\partial q_k} \delta q_k X_{ij} \quad (6)$$

Rearranging the order of summation:

$$\delta W = \sum_k \delta q_k \sum_i \sum_j \frac{\partial f_{ij}}{\partial q_k} X_{ij} \quad (7)$$

We define the generalised force associated with coordinate q_k :

$$F_k = \sum_i \sum_j \frac{\partial f_{ij}}{\partial q_k} X_{ij} \quad (8)$$

Each individual contribution $F_{ijk} = \frac{\partial f_{ij}}{\partial q_k} X_{ij}$ represents how much work the force X_{ij} does per unit virtual displacement of the generalised coordinate q_k , and F_k is the sum of all these contributions. A longer explanation and examples follow further below.

The virtual work could now also be expressed as ([Goldstein et al., 2002](#))

$$\delta W = \sum_k F_k \delta q_k = \mathbf{F} \cdot \delta \mathbf{q} \quad (9)$$

I.e., the dot product between the generalised force vector $\mathbf{F}$ and the vector of virtual displacements $\delta \mathbf{q}$.

Generalised force F_k in words The term $\frac{\partial f_{ij}}{\partial q_k}$ defines how much the degree of freedom ij in the original coordinate system virtually moves when the generalised coordinate k is virtually displaced. The displacement of a single generalised coordinate k can, in general, cause all degrees of freedom (i.e. every particle in every direction) to be displaced, which would cause all forces X_{ij} to do work. The generalised force F_k is the collection of all forces X_{ij} that perform work when q_k is displaced. This collection (sum) is weighted by $\frac{\partial f_{ij}}{\partial q_k}$ as to how much every force contributes to the work: If the derivative is large, particle i moves a lot in the j direction when q_k is virtually displaced. Hence, the force X_{ij} does proportionally much work.

Example, Generalised Coordinates and Force Think about a wind turbine blade with aerodynamic (distributed) loading. We might describe the deflection z_b of the blade by $z_b(t, x) = q(t) \cdot \Phi(x)$ with a generalised coordinate $q(t)$ and a mode shape $\Phi(x)$. This connects to two points stated above:

1. Reduction in degrees of freedom: Our beam has millions of degrees of freedom (it's made up of many particles) with many constraints between them. Writing down all of these degrees of freedom and their constraints is infeasible. However, with the generalised coordinate q (and the mode shape), we reduce the description of the displacement of our beam to just one degree of freedom.
2. Generalised forces: Let's reduce the distributed loading to a number of equivalent point forces X_i (dropping the j because we only work in z) that are distributed accordingly along the beam. If we now virtually displace q (i.e. change the value of q slightly), the whole beam displaces according to $\Phi(x)$. Hence, all X_i can, in general, do work even though just one degree of freedom was changed.

Exercise Go back to Lecture 8 to the example of three carts connected by springs. Use the generalised coordinates z_i (drop the j because there's only one dimension). Figure out what the $f_i(z_1, z_2, z_3)$ are. Then use Eq. (8) to derive the GF shown on the next lecture page.

3 Application: Beam Defined by Mode Shapes and Natural Frequencies

In this section, the use of modes with general coordinates is first described. Then, Lagrange’s equation is introduced with a small spoiler as to what it is good for. After that, a blade’s generalised mass matrix, stiffness matrix (assuming the blade as an Euler-Bernoulli beam), and stiffness matrix, and the generalised aerodynamic and gravitational forces are derived. In the end, a compact summary of the results is given.

3.1 Setup

Consider a beam (e.g. a wind turbine blade) with motion in two directions y and z . The beam has a known mass distribution $\rho A(x)$ along its length, three mode shapes and corresponding natural frequencies, but no explicit $EI(x)$ or $c(x)$ (damping) distributions. The three mode shapes in the beam’s coordinate system are:

- $\phi_1^y(x), \phi_1^z(x)$: first flapwise mode (components in y and z)
- $\phi_2^y(x), \phi_2^z(x)$: first edgewise mode (components in y and z)
- $\phi_3^y(x), \phi_3^z(x)$: second flapwise mode (components in y and z)

Each mode shape generally has components in both y and z of the global coordinate system when the global and blade y and z axes do not align (noticeable when, e.g., the blades are pitching).

The displacement field of the beam is approximated using these modes as shape functions. We may then introduce the generalised coordinates as the modal amplitudes $q_1(t), q_2(t), q_3(t)$:

$$\begin{aligned} y(x, t) &= \phi_1^y(x) q_1(t) + \phi_2^y(x) q_2(t) + \phi_3^y(x) q_3(t) \\ z(x, t) &= \phi_1^z(x) q_1(t) + \phi_2^z(x) q_2(t) + \phi_3^z(x) q_3(t) \end{aligned} \tag{10}$$

This is the connection to the general theory: $f_{ij}(q_k)$ of Eq. (1) is given by the mode shapes. For a point at position x along the beam, the “particle index” i corresponds to location x , and the “axis index” j is either y or z :

$$\begin{aligned} f_{x,y}(x, t) &= \phi_1^y(x) q_1(t) + \phi_2^y(x) q_2(t) + \phi_3^y(x) q_3(t) \\ f_{x,z}(x, t) &= \phi_1^z(x) q_1(t) + \phi_2^z(x) q_2(t) + \phi_3^z(x) q_3(t) \end{aligned} \tag{11}$$

Note that we treat the mode shapes as fixed functions of x – they do not depend on time or on the generalised coordinates. This implicitly assumes that the blade coordinate system is not rotating during the analysis (i.e. no pitching). If we simply calculate the mode shapes in the global coordinate system, new for each time step, we do it quasi-steadily.

3 APPLICATION: BEAM DEFINED BY MODE SHAPES AND NATURAL FREQUENCIES

The partial derivatives that appear in the generalised force definition Eq. (8) are simply the mode shapes:

$$\frac{\partial f_{x,y}}{\partial q_k} = \phi_k^y(x), \quad \frac{\partial f_{x,z}}{\partial q_k} = \phi_k^z(x), \quad k \in (1, 2, 3) \quad (12)$$

Where we have one value of k for each generalised coordinate.

3.2 Lagrange's Equation: The Governing Principle

The entire derivation that follows is built on Lagrange's equation for each generalised coordinate q_k . Hence, we compute kinetic and potential energies: Lagrange's equation transforms energy expressions into equations of motion.

For a system described by generalised coordinates q_k with kinetic energy T , potential energy V , and non-conservative generalised forces Q_k , Lagrange's equation states (Goldstein et al., 2002, § 1.4–1.6):

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_k} - \frac{\partial T}{\partial q_k} + \frac{\partial V}{\partial q_k} = Q_k \quad (13)$$

In there are our generalised mass and stiffness matrices. Some descriptions and spoilers that we will realise along the way:

- $\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_k}$: The time derivative of the derivative of the kinetic energy with respect to the generalised velocities. This will give us the generalised mass matrix.
- $\frac{\partial T}{\partial q_k}$: The derivative of the kinetic energy with respect to the generalised coordinates. This will vanish due to our choice of generalised coordinates.
- $\frac{\partial V}{\partial q_k}$: The derivative of the potential energy with respect to the generalised coordinates. This will give us the generalised stiffness matrix. We will use this in combination with our assumption of an Euler-Bernoulli beam as a blade to derive the connection between the generalised mass and stiffness matrices.
- Q_k : Generalised forces. These include non-conservative forces; those that cannot be accounted for by T or V . In our case, these are the aerodynamic forces. We will also include the gravitational forces despite them being conservative because their influence, as we will see, is time-dependent.

Since the Lagrangian framework does not naturally accommodate dissipative forces (such as structural damping), we can introduce a Rayleigh dissipation function $\mathcal{F}$ (Goldstein et al., 2002, § 1.5), which modifies Lagrange's equation to:

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_k} - \frac{\partial T}{\partial q_k} + \frac{\partial V}{\partial q_k} + \frac{\partial \mathcal{F}}{\partial \dot{q}_k} = Q_k \quad (14)$$

The term $\frac{\partial \mathcal{F}}{\partial \dot{q}_k}$ will give us our damping matrix.

In the following, we will derive the generalised mass, stiffness and damping matrices and the generalised forces for a beam (e.g. a wind turbine blade) under distributed aerodynamic and gravitational loading.

3.3 Generalised Mass Matrix

First, we write the kinetic energy of the beam in terms of our generalised coordinates. Then, we use Lagrange's equation to derive the generalised mass matrix.

Kinetic energy in terms of generalised coordinates The kinetic energy of the purely bending beam, neglecting rotational effects, is ([Géradin and Rixen, 2015](#), § 4.3.3):

$$T = \frac{1}{2} \int_0^L \rho A(x) [\dot{y}(x, t)^2 + \dot{z}(x, t)^2] dx \quad (15)$$

In other words, integrating the differential kinetic energy of each cross-section element of the beam. The velocities are obtained by differentiating the modal expansions (our $f_{ij} = f_{x,y}$) with respect to time. Since the mode shapes $\phi_k^y(x)$ and $\phi_k^z(x)$ are fixed functions of x and do not depend on time, only the generalised coordinates $q_k(t)$ are differentiated:

$$\dot{y} = \sum_k \phi_k^y(x) \dot{q}_k, \quad \dot{z} = \sum_k \phi_k^z(x) \dot{q}_k \quad (16)$$

Squaring these expressions (expanding the product of two sums):

$$\dot{y}^2 = \left(\sum_k \phi_k^y \dot{q}_k \right) \left(\sum_m \phi_m^y \dot{q}_m \right) = \sum_k \sum_m \phi_k^y \phi_m^y \dot{q}_k \dot{q}_m \quad (17)$$

$$\dot{z}^2 = \left(\sum_k \phi_k^z \dot{q}_k \right) \left(\sum_m \phi_m^z \dot{q}_m \right) = \sum_k \sum_m \phi_k^z \phi_m^z \dot{q}_k \dot{q}_m \quad (18)$$

This squaring might come as a surprise. Why not just $\dot{y}^2 \stackrel{?}{=} \sum_k (\phi_k^y(x) \dot{q}_k)^2$? Because it is a sum! Try the sum $y = a + b$ and now square it. The same is happening in the above expressions, and that's the reason we have to introduce the second index m .

Substituting into the kinetic energy expression:

$$T = \frac{1}{2} \int_0^L \rho A(x) \left[\sum_k \sum_m \phi_k^y \phi_m^y \dot{q}_k \dot{q}_m + \sum_k \sum_m \phi_k^z \phi_m^z \dot{q}_k \dot{q}_m \right] dx \quad (19)$$

Since $\dot{q}_k$ and $\dot{q}_m$ do not depend on x , they can be pulled outside the integral:

$$T = \frac{1}{2} \sum_k \sum_m \dot{q}_k \dot{q}_m \int_0^L \rho A(x) [\phi_k^y(x) \phi_m^y(x) + \phi_k^z(x) \phi_m^z(x)] dx \quad (20)$$

Next, if we define the factors M_{km} that make up a matrix $\mathbf{M}$:

$$M_{km} = \int_0^L \rho A(x) [\phi_k^y(x) \phi_m^y(x) + \phi_k^z(x) \phi_m^z(x)] dx \quad (21)$$

What we are left with is

$$T = \frac{1}{2} \sum_k \sum_m \dot{q}_k \dot{q}_m M_{km} \quad (22)$$

For now, M_{km} are simply some factors; we will see soon what they represent.

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Deriving the inertial term (mass) via Lagrange's equation We now carry out the differentiation prescribed by the first term in Lagrange's equation to show how the mass matrix enters the equations of motion, i.e. we carry out $\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_k}$.

Step 1: Differentiate T with respect to $\dot{q}_k$. We now differentiate T with respect to the generalised velocity $\dot{q}_k$. The k index hence chooses a specific generalised velocity. However, in Eq. (22) above, we used k as a summation index. Both represent something different. Hence, we rename the k from the kinetic energy expression to r . We also rename the kinetic energy expression's m to s . Not because we have to, but for stylistic reasons. Thus, starting from:

$$T = \frac{1}{2} \sum_r \sum_s M_{rs} \dot{q}_r \dot{q}_s \quad (23)$$

we differentiate with respect to $\dot{q}_k$. The product $\dot{q}_r \dot{q}_s$ depends on $\dot{q}_k$ only when $r = k$ or $s = k$ (or both). We can evaluate this derivative by expanding:

$$\frac{\partial}{\partial \dot{q}_k} (\dot{q}_r \dot{q}_s) = \delta_{rk} \dot{q}_s + \dot{q}_r \delta_{sk} \quad (24)$$

where δ_{rk} is the Kronecker delta ($\delta_{rk} = 1$ if $r = k$, and 0 otherwise). Therefore:

$$\frac{\partial T}{\partial \dot{q}_k} = \frac{1}{2} \sum_r \sum_s M_{rs} (\delta_{rk} \dot{q}_s + \dot{q}_r \delta_{sk}) \quad (25)$$

The first term, summing over r with the Kronecker delta δ_{rk} , collapses to $r = k$:

$$\frac{1}{2} \sum_r \sum_s M_{rs} \delta_{rk} \dot{q}_s = \frac{1}{2} \sum_s M_{ks} \dot{q}_s \quad (26)$$

The second term, summing over s with δ_{sk} , collapses to $s = k$:

$$\frac{1}{2} \sum_r \sum_s M_{rs} \dot{q}_r \delta_{sk} = \frac{1}{2} \sum_r M_{rk} \dot{q}_r \quad (27)$$

Since the mass matrix is symmetric ($M_{rk} = M_{kr}$, which follows from the definition since $\phi_k^y(x) \phi_m^y(x)$ and $\phi_m^y(x) \phi_k^y(x)$ are just real numbers, and $a \cdot b = b \cdot a$), we can relabel the dummy indices $s \rightarrow m$ in the first sum and $r \rightarrow m$ in the second to combine them:

$$\frac{\partial T}{\partial \dot{q}_k} = \frac{1}{2} \sum_m M_{km} \dot{q}_m + \frac{1}{2} \sum_m M_{km} \dot{q}_m = \sum_m M_{km} \dot{q}_m \quad (28)$$

Step 2: Differentiate with respect to time. Since the entries M_{km} are constants (they depend only on the fixed mode shapes and mass distribution, not on time), the time derivative passes straight through to $\dot{q}_m$:

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_k} = \sum_m M_{km} \ddot{q}_m \quad (29)$$

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And now we see: the matrix $\mathbf{M}$ multiplies the generalised accelerations, and no other term of our original Lagrange equation yields acceleration terms. In our Lagrange equation, we have the generalised forces on the right side and now the $\sum_m M_{km} \ddot{q}_m$ on the left side. Referring to Newton's 2nd law of mass times acceleration equals force, we see that our M_{km} builds the matrix that we refer to as the mass matrix by Newton's definition. Hence, M_{km} are the components of what we call the generalised mass matrix $\mathbf{M}$.

Rewriting our Lagrange equation in matrix form, we would have

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\mathbf{q}}} = \mathbf{M} \ddot{\mathbf{q}} \quad (30)$$

Mode orthogonality and the diagonal mass matrix

If the modes are orthogonal with respect to the mass distribution, then $M_{km} = 0$ for $k \neq m$, and the mass matrix is diagonal:

$$M_{kk} = \int_0^L \rho A(x) [(\phi_k^y)^2 + (\phi_k^z)^2] dx \quad (31)$$

For our modal approach, this orthogonality is not an assumption but a consequence of the physics ([Meirovitch, 1967](#), § 5.6).

3.4 Generalised Stiffness Matrix

Now we turn to the generalised stiffness matrix. Instead of the kinetic energy, we have to look at the potential energy. This should not come as a complete surprise; if one bends a beam, one overcomes its stiffness, and while doing so, one stores potential energy in the beam.

Potential energy in terms of generalised coordinates In the standard Lagrangian framework, the stiffness matrix arises from the elastic strain (potential) energy. For an Euler-Bernoulli beam purely bending around two principal axes, the strain energy is ([Langhaar, 1962](#), p. 40):

$$V = \frac{1}{2} \int_0^L EI_z(x) y''(x, t)^2 dx + \frac{1}{2} \int_0^L EI_y(x) z''(x, t)^2 dx \quad (32)$$

where $EI_z(x)$ is the bending stiffness about the z -axis (which resists curvature in the y -direction), $EI_y(x)$ is the bending stiffness about the y -axis (which resists curvature in the z -direction), and primes denote spatial derivatives ($y'' = \partial^2 y / \partial x^2$).

Substituting the modal expansion, the curvatures are:

$$y''(x, t) = \sum_k \phi_k^{y''}(x) q_k(t), \quad z''(x, t) = \sum_k \phi_k^{z''}(x) q_k(t) \quad (33)$$

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where $\phi_k^{y''} = \frac{d^2 \phi_k^y}{dx^2}$ is the second spatial derivative of the mode shape. Squaring:

$$(y'')^2 = \sum_k \sum_m \phi_k^{y''} \phi_m^{y''} q_k q_m, \quad (z'')^2 = \sum_k \sum_m \phi_k^{z''} \phi_m^{z''} q_k q_m \quad (34)$$

Substituting into the strain energy:

$$V = \frac{1}{2} \sum_k \sum_m q_k q_m \int_0^L [EI_z(x) \phi_k^{y''} \phi_m^{y''} + EI_y(x) \phi_k^{z''} \phi_m^{z''}] dx \quad (35)$$

If we now define the factors K_{km}

$$K_{km} = \int_0^L [EI_z(x) \phi_k^{y''}(x) \phi_m^{y''}(x) + EI_y(x) \phi_k^{z''}(x) \phi_m^{z''}(x)] dx \quad (36)$$

Then we are left with

$$V = \frac{1}{2} \sum_k \sum_m q_k q_m K_{km} \quad (37)$$

And again, for now, these are simply factors; we will see soon what they represent.

Deriving the stiffness term via Lagrange's equation Applying Lagrange's equation, we differentiate V with respect to q_k :

$$\frac{\partial V}{\partial q_k} = \frac{\partial}{\partial q_k} \left(\frac{1}{2} \sum_r \sum_s K_{rs} q_r q_s \right) \quad (38)$$

The derivative follows the same pattern as for the kinetic energy. Using $\frac{\partial}{\partial q_k}(q_r q_s) = \delta_{rk} q_s + q_r \delta_{sk}$:

$$\frac{\partial V}{\partial q_k} = \frac{1}{2} \sum_r \sum_s K_{rs} (\delta_{rk} q_s + q_r \delta_{sk}) = \frac{1}{2} \sum_s K_{ks} q_s + \frac{1}{2} \sum_r K_{rk} q_r \quad (39)$$

Since the stiffness matrix is symmetric ($K_{rk} = K_{kr}$, which follows from the symmetry of the integrand again), the two sums are equal, and:

$$\frac{\partial V}{\partial q_k} = \sum_m K_{km} q_m \quad (40)$$

And now we see: the matrix $\mathbf{K}$ multiplies the generalised displacements, and no other term of our original Lagrange equation yields displacement terms. Just as Hooke's law states $F = \kappa x$ (force equals stiffness times displacement), here $\frac{\partial V}{\partial q_k} = \sum_m K_{km} q_m$ states that the elastic restoring force on mode k is the stiffness matrix row k times the displacement vector. In matrix form, we have the generalised stiffness matrix $\mathbf{K}$

$$\frac{\partial V}{\partial q_k} = \mathbf{K} \mathbf{q} \quad (41)$$

The practical problem: $EI(x)$ is not available

The potential energy route above requires knowledge of $EI(x)$ to evaluate the stiffness integrals. In many practical situations, the bending stiffness distribution is not readily available, but the natural frequencies ω_k are known (from measurements, finite-element analysis, or manufacturer data). In this case, we can determine the stiffness matrix entries without $EI(x)$ by exploiting the eigenvalue problem.

Generalised stiffness via the eigenvalue problem

Each mode shape ϕ_k and its natural frequency ω_k satisfy the free-vibration eigenvalue problem. Assuming no structural coupling between the modes, the eigenvalue problem decouples into two independent equations, one for each bending direction (Rao, 2017, § 8.5.1):

$$\frac{d^2}{dx^2} (EI_z(x) \phi_k^{y''}(x)) = \omega_k^2 \rho A(x) \phi_k^y(x) \quad (42)$$

$$\frac{d^2}{dx^2} (EI_y(x) \phi_k^{z''}(x)) = \omega_k^2 \rho A(x) \phi_k^z(x) \quad (43)$$

To connect the stiffness and mass matrices, we treat each direction independently and then add the results. Multiply Eq. (42) by $\phi_m^y(x)$ and integrate over the beam length:

$$\int_0^L \phi_m^y \frac{d^2}{dx^2} (EI_z \phi_k^{y''}) dx = \omega_k^2 \int_0^L \rho A \phi_k^y \phi_m^y dx \quad (44)$$

Similarly, multiply Eq. (43) by $\phi_m^z(x)$ and integrate:

$$\int_0^L \phi_m^z \frac{d^2}{dx^2} (EI_y \phi_k^{z''}) dx = \omega_k^2 \int_0^L \rho A \phi_k^z \phi_m^z dx \quad (45)$$

The right-hand sides, when adding Eqs. (44) and (45), give $\omega_k^2 M_{km}$ by definition of the mass matrix. Each left-hand side almost has the quantities we need for the generalised stiffness – the derivatives are not yet in the right form. We apply integration by parts twice to each left-hand side to move derivatives from the $EI \phi_k''$ term onto ϕ_m . The procedure is identical for both directions, so we show it once for the y -direction in Section A. What we find, after summing both directions, is

$$\underbrace{\int_0^L EI_z \phi_k^{y''} \phi_m^{y''} dx + \int_0^L EI_y \phi_k^{z''} \phi_m^{z''} dx}_{K_{km}} = \omega_k^2 \underbrace{\int_0^L \rho A (\phi_k^y \phi_m^y + \phi_k^z \phi_m^z) dx}_{M_{km}} \quad (46)$$

That is:

$$K_{km} = \omega_k^2 M_{km} \quad (47)$$

Using the eigenvalue approach, we avoid the need to know $EI(x)$. Note that because the bending directions are decoupled, the result is simply the sum of two independent scalar eigenvalue arguments — one for y and one for z .

Now applying orthogonality ($M_{km} = 0$ for $k \neq m$):

$$K_{km} = \omega_k^2 \cdot 0 = 0 \quad \text{for } k \neq m \quad (48)$$

Both matrices are diagonal, and the diagonal entries satisfy:

$$\boxed{K_{kk} = \omega_k^2 M_{kk}} \quad (49)$$

3.5 Generalised Damping Matrix

Why damping sits outside the Lagrangian

The Lagrangian framework is built on the difference between kinetic and potential energy, and it naturally handles conservative systems. Structural damping, however, is a dissipative (non-conservative) phenomenon: it removes energy from the system as heat, and there is no potential energy function whose gradient gives the damping force. Therefore, damping cannot be captured by T or V alone.

To include damping within a Lagrangian-like framework, one introduces the Rayleigh dissipation function $\mathcal{F}$ (Goldstein et al., 2002, § 1.5), defined as

$$\mathcal{F} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{C} \dot{\mathbf{q}} = \frac{1}{2} \sum_k \sum_m C_{km} \dot{q}_k \dot{q}_m \quad (50)$$

Differentiating $\mathcal{F}$ with respect to $\dot{q}_k$ follows the same pattern as the earlier differentiations:

$$\frac{\partial \mathcal{F}}{\partial \dot{q}_k} = \sum_m C_{km} \dot{q}_m \quad (51)$$

With similar reasoning as for $\mathbf{M}$ and $\mathbf{K}$, this gives the damping force term $\mathbf{C} \dot{\mathbf{q}}$ in the equation of motion.

Determining the damping coefficients via the Single Degree of Freedom analogy

Since the modes are orthogonal, the mass and stiffness matrices are both diagonal. If we further assume that damping does not couple the modes – that is, the damping matrix is also diagonal in the modal basis – then each mode behaves as an independent single-degree-of-freedom (SDOF) oscillator:

$$M_{kk} \ddot{q}_k + C_{kk} \dot{q}_k + K_{kk} q_k = Q_k \quad (52)$$

This assumption is called modal damping (Géradin and Rixen, 2015, § 3.1.2): each mode has its own damping coefficient C_{kk} , and there are no off-diagonal damping terms that would couple one mode's velocity to another mode's equation of motion. It is not a consequence of physics but is rather physically reasonable for lightly damped structures (like wind turbine blades), where damping is small enough that it does not significantly alter the mode shapes or introduce coupling between them.

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For an SDOF system with mass m , stiffness κ , natural frequency $\omega = \sqrt{\kappa/m}$, and damping ratio ζ , the damping coefficient is (Rao, 2017, § 2.6.2):

$$c = 2\zeta\omega m \quad (53)$$

Applying this to each mode with generalised mass M_{kk} and natural frequency ω_k :

$$C_{kk} = 2\zeta_k \omega_k M_{kk} \quad (54)$$

The damping ratio ζ_k can be specified directly, or equivalently, through the logarithmic decrement δ_k . The logarithmic decrement is the natural logarithm of the ratio of successive peak amplitudes in a free-decay test: $\delta_k = \ln(q_{\text{peak},n}/q_{\text{peak},n+1})$. It is related to the damping ratio by (Rao, 2017, § 2.6.3):

$$\delta_k = \frac{2\pi\zeta_k}{\sqrt{1-\zeta_k^2}} \approx 2\pi\zeta_k \quad \text{for small } \zeta_k \quad (55)$$

The approximation holds because $\sqrt{1-\zeta_k^2} \approx 1$ when $\zeta_k \ll 1$. Solving for ζ_k :

$$\zeta_k \approx \frac{\delta_k}{2\pi} \quad (56)$$

Substituting into the expression for C_{kk} :

$$C_{kk} = 2 \cdot \frac{\delta_k}{2\pi} \cdot \omega_k \cdot M_{kk} = \frac{\delta_k}{\pi} \omega_k M_{kk} \quad (57)$$

$$C_{kk} = \frac{\delta_k}{\pi} \omega_k M_{kk}$$

(58)

3.6 Generalised Forces from Distributed and Gravitational Loads

The generalised forces Q_k are precisely the terms that appear on the right-hand side of Lagrange's equation. They represent the work done by external applied forces that are not already accounted for by the potential energy V or the dissipation function $\mathcal{F}$. In our formulation, the aerodynamic loads are non-conservative (they depend on the flow field, not on a stored energy) and are not velocity-dependent dissipation, so they must enter as generalised forces. Gravity is conservative in principle, but its projection onto the rotating blade coordinates is time-dependent, making it more practical to treat it as an external generalised force.

Aerodynamic loads

Consider a distributed aerodynamic load on the beam with components $p_y^{\text{aero}}(x)$ and $p_z^{\text{aero}}(x)$ (force per unit length) in the blade's local y and z directions. The virtual work done by this distributed load is:

$$\delta W^{\text{aero}} = \int_0^L (p_y^{\text{aero}}(x) \delta y + p_z^{\text{aero}}(x) \delta z) dx \quad (59)$$

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Where we displace the whole blade by the same δy and δz , so every blade element force $p_y^{\text{aero}} dx$ and $p_z^{\text{aero}} dx$ are displaced by the same δy and δz , respectively.

Substituting the modal expansions for the virtual displacements:

$$\delta y = \sum_k \phi_k^y(x) \delta q_k, \quad \delta z = \sum_k \phi_k^z(x) \delta q_k \quad (60)$$

We get

$$\delta W^{\text{aero}} = \int_0^L \left(p_y^{\text{aero}} \sum_k \phi_k^y \delta q_k + p_z^{\text{aero}} \sum_k \phi_k^z \delta q_k \right) dx \quad (61)$$

Exchanging the order of summation and integration (valid because δq_k does not depend on x):

$$\delta W^{\text{aero}} = \sum_k \delta q_k \int_0^L \left(p_y^{\text{aero}}(x) \phi_k^y(x) + p_z^{\text{aero}}(x) \phi_k^z(x) \right) dx \quad (62)$$

Since the generalised force is defined by $\delta W = \sum_k Q_k \delta q_k$, we read off the aerodynamic generalised force for coordinate q_k :

$$Q_k^{\text{aero}} = \int_0^L \left(p_y^{\text{aero}}(x) \phi_k^y(x) + p_z^{\text{aero}}(x) \phi_k^z(x) \right) dx \quad (63)$$

Gravitational loads

Gravity acts as a distributed body force with magnitude $\rho A(x) g$ per unit length, pointing vertically downward. For a wind turbine blade, this vertical force must be projected onto the blade's local y and z axes. This projection depends on the rotor azimuth angle $\psi(t)$ and the pitch angle, so the gravity components in blade coordinates are time-varying.

Let $g_y(x, t)$ and $g_z(x, t)$ be the components of gravitational acceleration resolved into the blade's local y and z directions. The gravitational load per unit length is:

$$p_y^{\text{grav}}(x, t) = \rho A(x) g_y(t), \quad p_z^{\text{grav}}(x, t) = \rho A(x) g_z(t) \quad (64)$$

The gravitational constant g is, of course, not variable in the global coordinate system, but its projection into the blade coordinate system is.

The gravitational generalised force is obtained by the same projection as for the aerodynamic loads:

$$Q_k^{\text{grav}} = \int_0^L \rho A(x) \left(g_y(x, t) \phi_k^y(x) + g_z(x, t) \phi_k^z(x) \right) dx \quad (65)$$

The gravitational force has a distinctive feature: since $\rho A(x)$ and the mode shapes are fixed properties of the blade, the only time-varying part is the projection of gravity through g_y and g_z , which depends on the rotor azimuth angle $\psi(t)$.

Total generalised force

The total generalised force is the sum of the aerodynamic and gravitational contributions:

$$Q_k = \int_0^L [(p_y^{\text{aero}} + \rho A g_y) \phi_k^y + (p_z^{\text{aero}} + \rho A g_z) \phi_k^z] dx \quad (66)$$

In words: the distributed load in each direction (aerodynamic plus gravitational) is projected onto each mode shape by integration. Each mode “picks up” the component of the total load that does work through that mode’s pattern of deformation.

3.7 Summary: Equation of Motion for the Beam

Collecting all terms from Lagrange’s equation:

$$\underbrace{\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_k}}_{\sum_m M_{km} \ddot{q}_m} - \underbrace{\frac{\partial T}{\partial q_k}}_0 + \underbrace{\frac{\partial V}{\partial q_k}}_{\sum_m K_{km} q_m} + \underbrace{\frac{\partial \mathcal{F}}{\partial \dot{q}_k}}_{\sum_m C_{km} \dot{q}_m} = Q_k \quad (67)$$

the full equation of motion for the three generalised coordinates is:

$$\mathbf{M} \ddot{\mathbf{q}} + \mathbf{C} \dot{\mathbf{q}} + \mathbf{K} \mathbf{q} = \mathbf{Q} \quad (68)$$

where all matrices are 3×3 and diagonal (for orthogonal modes with modal damping), with entries:

$$M_{kk} = \int_0^L \rho A(x) [(\phi_k^y)^2 + (\phi_k^z)^2] dx \quad (69)$$

$$K_{kk} = \omega_k^2 M_{kk} \quad (70)$$

$$C_{kk} = \frac{\delta_k}{\pi} \omega_k M_{kk} \quad (71)$$

$$Q_k = Q_k^{\text{aero}} + Q_k^{\text{grav}} \quad (72)$$

$$= \int_0^L (p_y^{\text{aero}} \phi_k^y + p_z^{\text{aero}} \phi_k^z) dx + \int_0^L \rho A(x) (g_y \phi_k^y + g_z \phi_k^z) dx \quad (73)$$

The chain of logic: Lagrange’s equation provides the framework that connects energies to equations of motion. The mode shapes and mass distribution give $\mathbf{M}$ through integration (from the kinetic energy T). The stiffness matrix $\mathbf{K}$ would normally come from the strain energy V , but since $EI(x)$ is unavailable, it is instead obtained from the natural frequencies via the eigenvalue relation $K_{kk} = \omega_k^2 M_{kk}$. Damping is introduced through the Rayleigh dissipation function $\mathcal{F}$, with the damping coefficients determined by the logarithmic decrement (or equivalently, the damping ratio) using the SDOF analogy. The distributed aerodynamic loads and the gravitational body force give $\mathbf{Q}$ through projection onto the modes via the principle of virtual work. Gravity enters only as an external force (it does not affect $\mathbf{M}$, $\mathbf{K}$, or $\mathbf{C}$). Each mode behaves as an independent damped oscillator driven by its own generalised force.

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Caution Do not treat the derived equations as general solutions to a beam with distributed loading. There are quite a few assumptions (also some that are not explicitly emphasised) taken in this derivation that limit the extent this document takes while also limiting the generality of the equations.

4 Full Wind Turbine Model: Tower, Shaft, and Three Flexible Blades

This section extends Section 3 to a whole (simplified) wind turbine. First, the degrees of freedom are defined. Then, the kinetic energy of the turbine is derived. Based on that, the generalised mass, stiffness, and damping matrices are derived. The last derivations are the generalised forces for the tower and shaft motion; those for the beams are the same as derived in Section 3. Physical interpretations are given for the components of the generalised mass matrix.

4.1 System Description

Now we turn to the derivation of a complete (simplified) wind turbine model with three coupled subsystems:

- **Tower:** Modelled as a single degree of freedom in the fore-aft direction (along the shaft axis). The tower top is connected to a linear spring with stiffness K_t , representing the first fore-aft bending mode of the tower. The effective tower-top mass M_N includes the nacelle, hub, and any non-rotating mass lumped at the tower top, but does *not* include the blade masses (these enter through the blade kinetic energy).
- **Shaft:** Rigid in torsion, but free to rotate. The rotational speed is governed by the balance between aerodynamic torque from the rotor and resistive torque from the generator. The shaft (plus generator and hub) has a rotational inertia I_s about the shaft axis, again excluding the blade contributions. In our course, we neglect I_s , but it is kept in here.
- **Blades:** Three identical blades, each modelled by three mode shapes and corresponding natural frequencies, as derived in Section 3. Each blade has its own set of three generalised coordinates (modal amplitudes). The blades are equally spaced at 120° azimuthal intervals.

4.2 Generalised Coordinates

The full system has $2 + 3 \times 3 = 11$ generalised coordinates:

$$\mathbf{q} = \left[x_t, \psi, q_1^{(1)}, q_2^{(1)}, q_3^{(1)}, q_1^{(2)}, q_2^{(2)}, q_3^{(2)}, q_1^{(3)}, q_2^{(3)}, q_3^{(3)} \right]^T \quad (74)$$

where:

- $x_t(t)$: tower-top fore-aft displacement
- $\psi(t)$: shaft rotation angle (rotor azimuth)
- $q_k^{(b)}(t)$: amplitude of mode k ($k = 1, 2, 3$) on blade b ($b = 1, 2, 3$)

The azimuth angle of blade b is:

$$\psi_b(t) = \psi(t) + \frac{2\pi(b-1)}{3} \quad (75)$$

so that the three blades are equally spaced around the rotor disc.

4.3 Velocity of a Point on Blade b

In Section 3, we derived that the damping and stiffness matrices of the beam can be directly related to its mass matrix, which itself is based on the kinetic energy. Then again, the kinetic energy is based on the velocity, which is therefore at the centre of the derivation of the equations of motion.¹

Every coupling term in the mass matrix will originate from the fact that a single physical point on the blade moves due to *three* causes simultaneously: tower translation, shaft rotation, and elastic deformation. We must express the total velocity of every blade point in terms of the generalised velocities before computing the kinetic energy.

Coordinate system

We now assume that the rotor plane is perpendicular to the shaft and that the tower motion is small. We have:

- **Fore-aft direction** (along the shaft axis, pointing downwind): this is the direction of tower-top motion x_t , and it is also the z (out-of-rotor-plane) direction for every blade, regardless of azimuth.
- **Tangential direction** (in the rotor plane, perpendicular to the blade span): this is the direction of velocity induced by shaft rotation, and it is also the y (in-plane) direction for each blade.

z velocity

When the tower top moves fore-aft at velocity $\dot{x}_t$, the hub translates, and every point on every blade is carried along in the z direction. This contribution is uniform along the blade span – a point at station x moves with the same fore-aft velocity as the hub (kinematically, not necessarily dynamically as described next). The elastic z deformation adds $\sum_k \phi_k^z(x) \dot{q}_k^{(b)}$ on top of this. Therefore, the total z velocity of a point at station x on blade b is:

$$v_z^{(b)}(x, t) = \dot{x}_t + \sum_{k=1}^3 \phi_k^z(x) \dot{q}_k^{(b)} \quad (76)$$

Note that this expression is the same for all three blades (the tower motion $\dot{x}_t$ projects equally onto every blade's z direction). The only difference between blades is which set of modal coordinates $q_k^{(b)}$ appears.

¹The damping and stiffness components for the tower and shaft motion are much less involved and will also be shown.

y velocity

When the shaft rotates at rate $\dot{\psi}$, a point at radial distance x from the hub traces a circle of radius x and has tangential (y) speed $x\dot{\psi}$. The elastic y deformation adds $\sum_k \phi_k^y(x) \dot{q}_k^{(b)}$. Therefore:

$$v_y^{(b)}(x, t) = x\dot{\psi} + \sum_{k=1}^3 \phi_k^y(x) \dot{q}_k^{(b)} \quad (77)$$

Again, the same form for all three blades (shaft rotation produces the same tangential velocity structure for every blade), with different modal coordinates.

4.4 Kinetic Energy of the Full System

The total kinetic energy has three contributions:

$$T = \underbrace{\frac{1}{2}M_N \dot{x}_t^2}_{\text{tower}} + \underbrace{\frac{1}{2}I_s \dot{\psi}^2}_{\text{shaft}} + \underbrace{\sum_{b=1}^3 T_{\text{blade}}^{(b)}}_{\text{three blades}} \quad (78)$$

With the tower/nacelle/hub equivalent mass M_N , the shaft rotational inertia I_s , and the kinetic energy of blade b based on Eq. (15). The first two terms of Eq. (78) are fine as they are, but the blade's kinetic energy is dependent on the squares of the velocities of Eqs. (76) and (77). Those we expand now.

Expanding $(v_z^{(b)})^2$

$$(v_z^{(b)})^2 = \left(\dot{x}_t + \sum_k \phi_k^z \dot{q}_k^{(b)} \right)^2 \quad (79)$$

Expanding the square using $(a + b)^2 = a^2 + 2ab + b^2$, where $a = \dot{x}_t$ and $b = \sum_k \phi_k^z \dot{q}_k^{(b)}$:

$$(v_z^{(b)})^2 = \dot{x}_t^2 + 2\dot{x}_t \sum_{k=1}^3 \phi_k^z(x) \dot{q}_k^{(b)} + \left(\sum_{k=1}^3 \phi_k^z(x) \dot{q}_k^{(b)} \right)^2 \quad (80)$$

The last term is the square of a sum, which expands as a double sum:

$$\left(\sum_{k=1}^3 \phi_k^z \dot{q}_k^{(b)} \right)^2 = \sum_{k=1}^3 \sum_{m=1}^3 \phi_k^z(x) \phi_m^z(x) \dot{q}_k^{(b)} \dot{q}_m^{(b)} \quad (81)$$

So:

$$(v_z^{(b)})^2 = \underbrace{\dot{x}_t^2}_{(A)} + \underbrace{2\dot{x}_t \sum_k \phi_k^z \dot{q}_k^{(b)}}_{(B)} + \underbrace{\sum_k \sum_m \phi_k^z \phi_m^z \dot{q}_k^{(b)} \dot{q}_m^{(b)}}_{(C)} \quad (82)$$

Expanding $\left(v_y^{(b)}\right)^2$

By the same scheme as for $\left(v_z^{(b)}\right)^2$, we get

$$\left(v_y^{(b)}\right)^2 = \left(x \dot{\psi} + \sum_k \phi_k^y \dot{q}_k^{(b)}\right)^2 = \underbrace{x^2 \dot{\psi}^2}_{(D)} + \underbrace{2x \dot{\psi} \sum_k \phi_k^y \dot{q}_k^{(b)}}_{(E)} + \underbrace{\sum_k \sum_m \phi_k^y \phi_m^y \dot{q}_k^{(b)} \dot{q}_m^{(b)}}_{(F)} \quad (83)$$

Sorting terms by velocity pair

Before integrating, it is essential to recognise which pair of generalised velocities each term contains, because different pairs contribute to different entries of the mass matrix. Remember: $T = \frac{1}{2} \sum_i \sum_j M_{ij} \dot{q}_i \dot{q}_j$. Based on the total kinetic energy from Eq. (78) and our expression for the squared velocities Eqs. (82) and (83), we get:

Term	Contains velocities	Contributes to
(A): $\dot{x}_t^2$	$\dot{x}_t \cdot \dot{x}_t$	M_{x_t, x_t} (tower–tower)
(B): $2 \dot{x}_t \phi_k^z \dot{q}_k^{(b)}$	$\dot{x}_t \cdot \dot{q}_k^{(b)}$	$M_{x_t, q_k^{(b)}}$ (tower–blade coupling)
(C): $\phi_k^z \phi_m^z \dot{q}_k^{(b)} \dot{q}_m^{(b)}$	$\dot{q}_k^{(b)} \cdot \dot{q}_m^{(b)}$	$M_{q_k^{(b)}, q_m^{(b)}}$ (blade–blade, same blade)
(D): $x^2 \dot{\psi}^2$	$\dot{\psi} \cdot \dot{\psi}$	$M_{\psi, \psi}$ (shaft–shaft)
(E): $2x \dot{\psi} \phi_k^y \dot{q}_k^{(b)}$	$\dot{\psi} \cdot \dot{q}_k^{(b)}$	$M_{\psi, q_k^{(b)}}$ (shaft–blade coupling)
(F): $\phi_k^y \phi_m^y \dot{q}_k^{(b)} \dot{q}_m^{(b)}$	$\dot{q}_k^{(b)} \cdot \dot{q}_m^{(b)}$	$M_{q_k^{(b)}, q_m^{(b)}}$ (blade–blade, same blade)

Notice that terms (C) and (F) both contribute to the blade–blade block – they combine to give the full generalised mass integral $\int m(\phi_k^z \phi_m^z + \phi_k^y \phi_m^y) dx$ that was derived in Section 3. Also notice that there are no terms containing $\dot{q}_k^{(b)} \dot{q}_m^{(b')}$ with $b \neq b'$: different blades do not cross-couple in the mass matrix, because each blade’s velocity depends only on its own modal coordinates.

Integrating over blade length and summing over blades

We now integrate each blade velocity-squared term over $[0, R]$ with weight distribution $m(x)$ and sum over the three blades to get the blades’ kinetic energy contribution. Since all blades are identical (same $m(x)$ and mode shapes), the integrals are the same for each blade – only the modal coordinates differ.

From term (A), summed over three blades:

$$\sum_{b=1}^3 \frac{1}{2} \int_0^R m(x) \dot{x}_t^2 dx = \frac{3}{2} \dot{x}_t^2 \int_0^R m(x) dx \quad (84)$$

From term (D), summed over three blades:

$$\sum_{b=1}^3 \frac{1}{2} \int_0^R m(x) x^2 \dot{\psi}^2 dx = \frac{3}{2} \dot{\psi}^2 \int_0^R m(x) x^2 dx \quad (85)$$

From term (B), for a single blade b :

$$\frac{1}{2} \int_0^R m(x) \cdot 2 \dot{x}_t \sum_k \phi_k^z \dot{q}_k^{(b)} dx = \dot{x}_t \sum_k \dot{q}_k^{(b)} \int_0^R m(x) \phi_k^z(x) dx \quad (86)$$

From term (E), for a single blade b :

$$\frac{1}{2} \int_0^R m(x) \cdot 2 x \dot{\psi} \sum_k \phi_k^y \dot{q}_k^{(b)} dx = \dot{\psi} \sum_k \dot{q}_k^{(b)} \int_0^R m(x) x \phi_k^y(x) dx \quad (87)$$

From terms (C) + (F), for a single blade b :

$$\frac{1}{2} \sum_k \sum_m \dot{q}_k^{(b)} \dot{q}_m^{(b)} \int_0^R m(x) [\phi_k^z \phi_m^z + \phi_k^y \phi_m^y] dx \quad (88)$$

Since our modes are orthogonal with respect to the mass distribution $m(x)$, the integral vanishes for $k \neq m$, leaving only the diagonal contributions:

$$\frac{1}{2} \sum_k \left(\dot{q}_k^{(b)} \right)^2 \int_0^R m(x) [(\phi_k^z)^2 + (\phi_k^y)^2] dx \quad (89)$$

The full kinetic energy assembled

Combining the tower and shaft terms with all blade contributions:

$$\begin{aligned} T = & \frac{1}{2} \left(M_N + 3 \int_0^R m dx \right) \dot{x}_t^2 + \frac{1}{2} \left(I_s + 3 \int_0^R m x^2 dx \right) \dot{\psi}^2 \\ & + \sum_{b=1}^3 \sum_{k=1}^3 \dot{x}_t \dot{q}_k^{(b)} \int_0^R m(x) \phi_k^z(x) dx \\ & + \sum_{b=1}^3 \sum_{k=1}^3 \dot{\psi} \dot{q}_k^{(b)} \int_0^R m(x) x \phi_k^y(x) dx \\ & + \frac{1}{2} \sum_{b=1}^3 \sum_k \left(\dot{q}_k^{(b)} \right)^2 \int_0^R m(x) [(\phi_k^z)^2 + (\phi_k^y)^2] dx \end{aligned} \quad (90)$$

This is a quadratic form in all 11 generalised velocities, from which every entry of the 11×11 mass matrix can be read off, which we will do next.

4.5 Generalised Mass Matrix

Before reading off the matrix components, one might have noticed that some terms in Eq. (90) have a factor of $1/2$ and others do not. To address this, we can expand the sum of the kinetic energy Eq. (22) into ii and ij terms:

$$T = \frac{1}{2} \sum_{i=1}^n M_{ii} \dot{q}_i^2 + \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^n M_{ij} \dot{q}_i \dot{q}_j \quad (91)$$

The off-diagonal terms ij (the second sum) double count: for example, the pair $\{1, 2\}$ contributes $M_{12} \dot{q}_1 \dot{q}_2$ and $M_{21} \dot{q}_2 \dot{q}_1$, which, when summed, give $2M_{12} \dot{q}_1 \dot{q}_2$ since $M_{12} = M_{21}$. Applying this, we can rewrite Eq. (91) as

$$T = \frac{1}{2} \sum_{i=1}^n M_{ii} \dot{q}_i^2 + \frac{1}{2} \cdot 2 \sum_{1 \leq i < j \leq n} M_{ij} \dot{q}_i \dot{q}_j \quad (92)$$

$$= \frac{1}{2} \sum_{i=1}^n M_{ii} \dot{q}_i^2 + 1 \sum_{1 \leq i < j \leq n} M_{ij} \dot{q}_i \dot{q}_j \quad (93)$$

In Eq. (93) we see that for M_{ii} components, we need to have a $1/2$ factor, while for M_{ij} components, we need to have a factor of 1. Looking at Eq. (90), this is exactly what we have already, and we can read off the components as they are.

Tower–tower: M_{x_t, x_t}

From the $\dot{x}_t^2$ terms (tower mass plus term (A) from all three blades):

$$T \supset \frac{1}{2} \left(M_N + 3 \int_0^R m(x) dx \right) \dot{x}_t^2 \quad (94)$$

Reading off the diagonal entry:

$$\boxed{M_{x_t, x_t} = M_N + 3 \int_0^R m(x) dx} \quad (95)$$

Physical interpretation When the tower moves fore-aft, it carries the nacelle mass plus all three blade masses. The factor of 3 arises because all three blades are carried equally by the tower.

Shaft–shaft: $M_{\psi, \psi}$

From the $\dot{\psi}^2$ terms (shaft inertia plus term (D) from all three blades):

$$T \supset \frac{1}{2} \left(I_s + 3 \int_0^R m(x) x^2 dx \right) \dot{\psi}^2 \quad (96)$$

$$\boxed{M_{\psi, \psi} = I_s + 3 \int_0^R m(x) x^2 dx} \quad (97)$$

Physical interpretation The total rotational inertia is the shaft inertia plus the rotational inertia of all three blades. For each blade, $\int m x^2 dx$ is the second moment of mass about the rotation axis (the moment arm is x , the radial distance from the hub).

Tower-blade mode: $M_{x_t, q_k^{(b)}}$

From term (B), the kinetic energy contains (for each blade b and mode k):

$$T \supset \dot{x}_t \dot{q}_k^{(b)} \int_0^R m(x) \phi_k^z(x) dx \quad (98)$$

Hence:

$$\boxed{M_{x_t, q_k^{(b)}} = \int_0^R m(x) \phi_k^z(x) dx} \quad (99)$$

Physical interpretation: This entry couples tower fore-aft acceleration to blade mode k . When the tower accelerates fore-aft, every point on the blade experiences an inertial load in the z direction. This load excites mode k to the extent that mode k has a z component. The coupling integral measures how much of the blade's mass participates in the z motion of mode k : if ϕ_k^z is large where m is large, the coupling is strong. For a purely y mode ($\phi_k^z = 0$ everywhere), the tower motion does not excite it at all, and the coupling vanishes.

Note that this coupling is the same for all three blades (same mass distribution, same mode shapes). What differs between blades is only the modal coordinate $q_k^{(b)}$ – the structural coupling integral is blade-independent.

We define a shorthand for this integral that will appear repeatedly:

$$S_k^z \equiv \int_0^R m(x) \phi_k^z(x) dx \quad (100)$$

Shaft-blade mode: $M_{\psi, q_k^{(b)}}$

From term (E), the kinetic energy contains (for each blade b and mode k):

$$T \supset \dot{\psi} \dot{q}_k^{(b)} \int_0^R m(x) x \phi_k^y(x) dx \quad (101)$$

$$\boxed{M_{\psi, q_k^{(b)}} = \int_0^R m(x) x \phi_k^y(x) dx} \quad (102)$$

Physical interpretation This entry couples shaft rotational acceleration to blade mode k . When the shaft accelerates, a point at radius x experiences a tangential inertial load proportional to x (the moment arm). This tangential load projects onto the y component of each mode. The coupling integral is a mass-weighted first moment of the y

4 FULL WIND TURBINE MODEL: TOWER, SHAFT, AND THREE FLEXIBLE BLADES

mode shape: it measures how much angular momentum is associated with mode k 's y motion.

The factor x (moment arm) distinguishes this coupling from the tower coupling. Tower motion applies a uniform inertial load along the blade; shaft rotation applies a load that increases linearly with distance from the hub.

Again, this is the same for all three blades. We define:

$$S_k^y \equiv \int_0^R m(x) x \phi_k^y(x) dx \quad (103)$$

Blade-blade, same blade: $M_{q_k^{(b)}, q_m^{(b)}}$

From terms (C) and (F) combined, for a single blade b :

$$T \supset \frac{1}{2} \left(\dot{q}_k^{(b)} \right)^2 \int_0^R m(x) \left[(\phi_k^z)^2 + (\phi_k^y)^2 \right] dx \quad (104)$$

$$\boxed{M_{q_k^{(b)}, q_k^{(b)}} = GM_k = \int_0^R m(x) \left[(\phi_k^y)^2 + (\phi_k^z)^2 \right] dx} \quad (105)$$

Again, the entries are the same for all three blades.

Structure of the full 11×11 mass matrix

Ordering the generalised coordinates as

$$\mathbf{q} = [x_t, \psi, q_1^{(1)}, q_2^{(1)}, q_3^{(1)}, q_1^{(2)}, q_2^{(2)}, q_3^{(2)}, q_1^{(3)}, q_2^{(3)}, q_3^{(3)}]^T \quad (106)$$

the mass matrix has the block structure:

$$\mathbf{M} = \begin{bmatrix} M_{x_t, x_t} & 0 & \mathbf{s}_z^T & \mathbf{s}_z^T & \mathbf{s}_z^T \\ 0 & M_{\psi, \psi} & \mathbf{s}_y^T & \mathbf{s}_y^T & \mathbf{s}_y^T \\ \mathbf{s}_z & \mathbf{s}_y & \mathbf{G} & \mathbf{0} & \mathbf{0} \\ \mathbf{s}_z & \mathbf{s}_y & \mathbf{0} & \mathbf{G} & \mathbf{0} \\ \mathbf{s}_z & \mathbf{s}_y & \mathbf{0} & \mathbf{0} & \mathbf{G} \end{bmatrix} \quad (107)$$

where:

- $\mathbf{s}_z = [S_1^z, S_2^z, S_3^z]^T$ is the vector of tower-blade coupling integrals
- $\mathbf{s}_y = [S_1^y, S_2^y, S_3^y]^T$ is the vector of shaft-blade coupling integrals
- $\mathbf{G} = \text{diag}(GM_1, GM_2, GM_3)$ is the diagonal matrix of generalised blade masses

That these entries are indeed the mass matrix, i.e. that Lagrange's equation produces exactly accelerations multiplied by these coefficients and nothing else, can be done as an exercise.

4.6 Generalised Stiffness Matrix

The stiffness matrix arises from differentiating the potential energy V with respect to the generalised coordinates. Each subsystem contributes independently to V , with no cross-stiffness terms.

Tower

The tower is modelled as a linear spring:

$$V_{\text{tower}} = \frac{1}{2} K_t x_t^2 \quad (108)$$

$$\frac{\partial V_{\text{tower}}}{\partial x_t} = K_t x_t \quad (109)$$

All other partial derivatives of V_{tower} are zero (the tower spring depends only on x_t).

Shaft

The shaft is free to rotate — there is no elastic restoring torque on ψ . The torque balance comes entirely from the generalised forces (aerodynamic torque minus generator torque), not from a stored potential energy. Therefore:

$$V_{\text{shaft}} = 0, \quad \frac{\partial V_{\text{shaft}}}{\partial \psi} = 0 \quad (110)$$

Blade modes

As derived in Section 3, the generalised stiffness for each mode is obtained from the eigenvalue relation:

$$\frac{\partial V_{\text{blade}}^{(b)}}{\partial q_k^{(b)}} = \sum_m K_{km} q_m^{(b)} = \omega_k^2 G M_k q_k^{(b)} \quad (111)$$

where the last equality uses mode orthogonality.

Cross-stiffness terms

There are no cross-stiffness terms between any pair of generalised coordinates from different subsystems. The tower spring depends only on x_t . The blade strain energy depends only on that blade's modal coordinates. The shaft has no elastic energy.

The full stiffness matrix

With $GB_i = \omega_i^2 G M_i$

$$\mathbf{K} = \text{diag}(K_t, 0, GB_1, GB_2, GB_3, GB_1, GB_2, GB_3, GB_1, GB_2, GB_3) \quad (112)$$

Again, the stiffness matrix is purely diagonal. All coupling in this system enters through the mass matrix (inertial coupling) and the generalised forces (aerodynamic and gravitational coupling), not through stiffness.

4.7 Generalised Damping Matrix

Following the Rayleigh dissipation function approach from Section 3, and assuming modal damping (no cross-coupling in damping).

Tower

If the tower has a damping ratio ζ_t associated with its fore-aft mode, the damping coefficient is:

$$C_{x_t, x_t} = 2 \zeta_t \omega_t M_{x_t, x_t} \quad (113)$$

where $\omega_t = \sqrt{K_t/M_{x_t, x_t}}$ is the tower fore-aft natural frequency.

Shaft

There is no structural damping associated with the shaft rotation (the shaft is rigid).

$$C_{\psi, \psi} = 0 \quad (114)$$

Blade modes

As derived in Section 3, using the logarithmic decrement δ_k :

$$C_{q_k^{(b)}, q_k^{(b)}} = \frac{\delta_k}{\pi} \omega_k G M_k \quad (115)$$

This is the same for all three blades.

The full damping matrix

$$\mathbf{C} = \text{diag}(C_{x_t, x_t}, 0, C_{11}, C_{22}, C_{33}, C_{11}, C_{22}, C_{33}, C_{11}, C_{22}, C_{33}) \quad (116)$$

where $C_{kk} = \frac{\delta_k}{\pi} \omega_k G M_k$ is the damping coefficient for mode k . As with the blade-only case, the diagonality of the damping matrix is an engineering assumption (modal damping), not a consequence of the eigenvalue problem. It is well justified for lightly damped structures.

4.8 Generalised Forces

The generalised force for each coordinate is obtained from the principle of virtual work: impose a virtual displacement δq_i on one coordinate while holding all others fixed, and compute the work done by all external forces.

Tower: Q_{x_t}

A virtual displacement δx_t moves the entire rotor (hub plus all three blades) in the fore-aft direction. Every point on every blade translates by δx_t in the z direction. The work done by the aerodynamic z loads is:

$$\delta W_{x_t}^{\text{aero}} = \sum_{b=1}^3 \int_0^R p_n^{\text{aero},(b)}(x) \delta x_t dx = \delta x_t \sum_{b=1}^3 \int_0^R p_n^{\text{aero},(b)}(x) dx \quad (117)$$

The integral $\int_0^R p_n^{\text{aero},(b)} dx$ is the total z (thrust) force on blade b . Summing over all three blades gives the total rotor thrust F_{thrust} . Therefore:

$$Q_{x_t}^{\text{aero}} = F_{\text{thrust}} = \sum_{b=1}^3 \int_0^R p_n^{\text{aero},(b)}(x) dx \quad (118)$$

Gravity also contributes if the tower deflection has a gravitational projection, but for small fore-aft displacements, this is typically neglected.

$$\boxed{Q_{x_t} = F_{\text{thrust}}} \quad (119)$$

Shaft: Q_ψ

A virtual rotation $\delta\psi$ rotates all three blades by $\delta\psi$ about the shaft axis. A point at radius x on any blade moves tangentially by $x \delta\psi$. The work done by the aerodynamic tangential (y) loads is:

$$\delta W_\psi^{\text{aero}} = \sum_{b=1}^3 \int_0^R p_t^{\text{aero},(b)}(x) \cdot x \delta\psi dx = \delta\psi \sum_{b=1}^3 \int_0^R x p_t^{\text{aero},(b)}(x) dx \quad (120)$$

The integral $\int_0^R x p_t^{\text{aero},(b)} dx$ is the aerodynamic torque contribution from blade b . Summing over all three blades gives the total aerodynamic torque T_{aero} .

The generator exerts a resistive torque T_{gen} opposing the rotation. A virtual rotation $\delta\psi$ at the shaft does negative work $-T_{\text{gen}} \delta\psi$ against the generator torque. Therefore:

$$\boxed{Q_\psi = T_{\text{aero}} - T_{\text{gen}} = \sum_{b=1}^3 \int_0^R x p_t^{\text{aero},(b)}(x) dx - T_{\text{gen}}} \quad (121)$$

Note that gravity also contributes a torque on individual blades, but for a three-bladed rotor with 120° spacing, the gravity torques of the three blades sum to zero at all azimuths.

Blade mode k on blade b : $Q_{q_k^{(b)}}$

A virtual displacement $\delta q_k^{(b)}$ deforms blade b according to mode shape k : the z displacement is $\phi_k^z(x) \delta q_k^{(b)}$ and the y displacement is $\phi_k^y(x) \delta q_k^{(b)}$. No other blade is affected (the virtual displacement is specific to blade b). Hence, Eq. (66) applies for each blade individually.

4.9 Summary: Full Equation of Motion

Collecting all terms from the modified Lagrange's equation applied to each generalised coordinate:

$$\mathbf{M} \ddot{\mathbf{q}} + \mathbf{C} \dot{\mathbf{q}} + \mathbf{K} \mathbf{q} = \mathbf{Q} \quad (122)$$

with the 11×11 matrices and 11×1 force vector as derived above. Written out for each subsystem:

Tower:

$$M_{x_t, x_t} \ddot{x}_t + \sum_{b=1}^3 \sum_{k=1}^3 S_k^z \ddot{q}_k^{(b)} + C_{x_t, x_t} \dot{x}_t + K_t x_t = F_{\text{thrust}} \quad (123)$$

Shaft:

$$M_{\psi, \psi} \ddot{\psi} + \sum_{b=1}^3 \sum_{k=1}^3 S_k^y \ddot{q}_k^{(b)} = T_{\text{aero}} - T_{\text{gen}} \quad (124)$$

Blade mode k on blade b (for each $k = 1, 2, 3$ and $b = 1, 2, 3$):

$$S_k^z \ddot{x}_t + S_k^y \ddot{\psi} + G M_k \ddot{q}_k^{(b)} + C_{kk} \dot{q}_k^{(b)} + \omega_k^2 G M_k q_k^{(b)} = Q_{q_k^{(b)}} \quad (125)$$

The coupling structure is now explicit: the tower equation contains the blade accelerations $\ddot{q}_k^{(b)}$ (through S_k^z), the shaft equation contains them (through S_k^y), and each blade equation contains the tower and shaft accelerations. This is inertial coupling: it means that accelerating the tower produces inertial loads on the blades, and vice versa.

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A Integration by Parts for the Stiffness Orthogonality

The following derivation follows [Meirovitch \(2001, §8.5, eqs. \(8.96\)–\(8.108\)\)](#).

We show that the left-hand side of Eq. (42), when multiplied by $\phi_m^y(x)$ and integrated over $[0, L]$, reduces to the symmetric stiffness integral of Eq. (46).

First integration by parts. We integrate $\int_0^L \phi_m^y \frac{d^2}{dx^2} (EI_{zz} \phi_k^{y''}) dx$ by parts with $u = \phi_m^y$ and $dv = \frac{d^2}{dx^2} (EI_{zz} \phi_k^{y''}) dx$:

$$\int_0^L \phi_m^y \frac{d^2}{dx^2} (EI_{zz} \phi_k^{y''}) dx = \left[\phi_m^y \frac{d}{dx} (EI_{zz} \phi_k^{y''}) \right]_0^L - \int_0^L \phi_m^{y'} \frac{d}{dx} (EI_{zz} \phi_k^{y''}) dx. \quad (126)$$

The boundary term involves the product of displacement ϕ_m^y and shear force $\frac{d}{dx} (EI_{zz} \phi_k^{y''})$. It vanishes at both ends for standard boundary conditions:

- *Clamped*: $\phi_m^y = 0$ (zero displacement), so the product is zero regardless of the shear force.
- *Free*: $\frac{d}{dx} (EI_{zz} \phi_k^{y''}) = 0$ (zero shear force), so the product is zero regardless of the displacement.
- *Pinned*: $\phi_m^y = 0$ (zero displacement), so the product is zero.

Second integration by parts. The remaining integral is integrated by parts again with $u = \phi_m^{y'}$ and $dv = \frac{d}{dx} (EI_{zz} \phi_k^{y''}) dx$:

$$- \int_0^L \phi_m^{y'} \frac{d}{dx} (EI_{zz} \phi_k^{y''}) dx = - \left[\phi_m^{y'} EI_{zz} \phi_k^{y''} \right]_0^L + \int_0^L \phi_m^{y''} EI_{zz} \phi_k^{y''} dx. \quad (127)$$

The boundary term involves slope $\phi_m^{y'}$ and bending moment $EI_{zz} \phi_k^{y''}$. This also vanishes:

- *Clamped*: $\phi_m^{y'} = 0$ (zero slope).
- *Free*: $EI_{zz} \phi_k^{y''} = 0$ (zero bending moment).
- *Pinned*: $EI_{zz} \phi_k^{y''} = 0$ (zero bending moment).

In each case, one factor in each boundary product is zero. With both boundary terms vanishing, the y -direction integral reduces to:

$$\int_0^L \phi_m^y \frac{d^2}{dx^2} (EI_{zz} \phi_k^{y''}) dx = \int_0^L EI_{zz} \phi_k^{y''} \phi_m^{y''} dx. \quad (128)$$

The z -direction follows identically:

$$\int_0^L \phi_m^z \frac{d^2}{dx^2} (EI_{yy} \phi_k^{z''}) dx = \int_0^L EI_{yy} \phi_k^{z''} \phi_m^{z''} dx. \quad (129)$$

Adding both results and recognising the definitions of K_{km} (Eq. (36)) and M_{km} (Eq. (21)) recovers Eq. (46):

$$\underbrace{\int_0^L EI_{zz} \phi_k^{y''} \phi_m^{y''} dx + \int_0^L EI_{yy} \phi_k^{z''} \phi_m^{z''} dx}_{K_{km}} = \omega_k^2 \underbrace{\int_0^L \rho A (\phi_k^y \phi_m^y + \phi_k^z \phi_m^z) dx}_{M_{km}}. \quad (130)$$

Note also that stiffness orthogonality ($K_{km} = 0$ for $k \neq m$) is not an independent assumption. Writing the eigenvalue equation with ω_m^2 , subtracting, and using distinct frequencies gives $(\omega_k^2 - \omega_m^2)M_{km} = 0$, which for $\omega_k \neq \omega_m$ requires $M_{km} = 0$, and consequently $K_{km} = 0$.